

Pizza Sales Data Analysis

Business Requirements Document (BRD) & SQL Specification Document

Project Title	Pizza Sales Business Intelligence	Data Source	pizza_sales.csv
Document Version	v1.0 (Final)	Database Engine	PostgreSQL 17 / pgAdmin 4

1. EXECUTIVE SUMMARY & BUSINESS OBJECTIVES

The **Pizza Sales Analysis project** is designed to evaluate transactional store data, derive key operational metrics, and uncover sales patterns to support management decision-making across pricing, inventory, and marketing strategies.

Core Objectives:

- Quantify financial performance using key KPIs (Revenue, AOV, Order Volume).
- Analyze ordering behaviors across daily, hourly, and monthly timeframes.
- Identify top-performing and low-performing menu items for inventory optimization.
- Deliver structured PostgreSQL queries for dashboard integration.

2. DATA SCHEMA & FIELD DEFINITIONS

The operational dataset contains customer order details mapped across the following schema:

Field Name	Data Type	Description
order_id	Integer	Unique identifier for each transaction order.
pizza_id	Integer	Unique identifier for each specific item in an order.
pizza_name	Varchar	Name and flavor profile of the pizza.
quantity	Integer	Number of units ordered in the line item.
total_price	Decimal	Monetary sales value for the line item.
order_date / order_time	Date / Time	Order timestamp for temporal analysis.
pizza_category	Varchar	Product category (Classic, Supreme, Veggie, Chicken).
pizza_size	Varchar	Size designation (S, M, L, XL, XXL).

3. KEY PERFORMANCE INDICATORS (KPI QUERIES)

1. Total Revenue

Calculates cumulative sales generated across all transactions.

```
SELECT SUM(total_price) AS Total_Revenue FROM pizza_sales;
```

2. Average Order Value (AOV)

Measures the average monetary spend per individual customer order.

```
SELECT SUM(total_price) / COUNT(DISTINCT order_id) AS Avg_Order_Value FROM pizza_sales;
```

3. Total Pizzas Sold

Sum of total pizza units sold across all order lines.

```
SELECT SUM(quantity) AS Total_Pizza_Sold FROM pizza_sales;
```

4. Total Orders Count

Total volume of unique customer orders placed.

```
SELECT COUNT(DISTINCT order_id) AS Total_orders FROM pizza_sales;
```

5. Average Pizzas Per Order

Calculates average item density per individual transaction.

```
SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2)) /  
           CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2)) AS DECIMAL(10,2))  
       AS Avg_Pizza_Per_order  
FROM pizza_sales;
```

4. ADVANCED TRENDS & MENU ANALYTICS

Daily Order Distribution

Groups order volume by day of the week to assist in staffing schedule planning.

```
SELECT TO_CHAR(TO_DATE(order_date, 'DD-MM-YYYY'), 'Day') AS order_day,  
       COUNT(DISTINCT order_id) AS Total_orders  
FROM pizza_sales  
GROUP BY TO_CHAR(TO_DATE(order_date, 'DD-MM-YYYY'), 'Day');
```

Monthly Sales Performance

Tracks seasonality patterns and peak revenue months.

```
SELECT TO_CHAR(TO_DATE(order_date, 'DD-MM-YYYY'), 'FMMonth') AS Month_Name,  
       COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales  
GROUP BY TO_CHAR(TO_DATE(order_date, 'DD-MM-YYYY'), 'FMMonth')  
ORDER BY Total_orders DESC;
```

Sales Share by Pizza Size (Quarter 1)

Evaluates revenue contribution percentage by pizza size designation.

```
SELECT pizza_size,  
       CAST(SUM(total_price) AS DECIMAL(10,2)) AS Total_Sales,  
       CAST(SUM(total_price) * 100 / (  
           SELECT SUM(total_price) FROM pizza_sales  
           WHERE EXTRACT(QUARTER FROM TO_DATE(order_date, 'DD-MM-YYYY')) = 1  
       ) AS DECIMAL(10,2)) AS PCT  
FROM pizza_sales  
WHERE EXTRACT(QUARTER FROM TO_DATE(order_date, 'DD-MM-YYYY')) = 1  
GROUP BY pizza_size  
ORDER BY PCT DESC;
```

Top 5 Best Sellers (by Revenue)

Identifies top revenue drivers for promotional focus.

```
SELECT pizza_name, SUM(total_price) AS Total_Revenue  
FROM pizza_sales  
GROUP BY pizza_name  
ORDER BY Total_Revenue DESC  
LIMIT 5;
```

Bottom 5 Lowest Sellers (by Orders)

Highlights menu items candidate for recipe adjustment or removal.

```
SELECT pizza_name, COUNT(DISTINCT order_id) AS Total_Orders  
FROM pizza_sales  
GROUP BY pizza_name  
ORDER BY Total_Orders ASC  
LIMIT 5;
```

5. BUSINESS INSIGHTS & RECOMMENDATIONS

- **Promotional Focus:** Prioritize top-performing items (Classic and Chicken categories) in marketing campaigns.
- **Menu Optimization:** Review bottom 5 performing items to reduce raw material waste and streamline kitchen prep.
- **Labor Allocation:** Align kitchen shift schedules with peak daily and hourly order volumes.
- **Inventory Control:** Adjust stock levels according to size-wise demand velocity.